

Firm's Problem

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Firms maximize profits by choosing # of producers n , taking as given market tightness x, θ and prices p, w .

(Equivalent - firm chooses # employees ℓ
 [b/c $\ell = [1 + \hat{\tau}(\theta)] n$]
 - firm chooses # vacancies $\hat{v}$
 [b/c $\hat{v} = \ell / \hat{q}(\theta)$])

Profits : Revenue - Cost
 = Revenue from sale of services - Wage bill

$$= p \cdot f(x) \cdot k - w \cdot \ell$$

price of service selling proba capacity nominal wage employees

$$= p \cdot f(x) \cdot a \cdot n^{\alpha} - w [1 + \hat{\tau}(\theta)] \cdot n$$

technology

matching wedge

producers

function of n :
 concave in n
 ($\alpha < 1$)

standard concave maximization problem

$$\max_{n > 0} p \cdot a \cdot f(x) n^{\alpha} - w [1 + \hat{\tau}(\theta)] n$$

↳ derivative.

FOC

↳ sufficient

condition for global maximum

$$p \cdot a \cdot f(x) \alpha n^{\alpha-1} = w [1 + \hat{\tau}(\theta)]$$

producer
 wage for recruits to get producer

MR from 1 producer:

MR $\times$ price $\times$ selling prob

MC of product:

cost of 1 extra producer

$$\Leftrightarrow n^{\alpha-1} = \frac{[1 + \hat{\tau}(\theta)]}{\alpha a f(x)} \frac{w}{p}$$

$$\Leftrightarrow n = \left[\frac{a \alpha f(x)}{[1 + \hat{\tau}(\theta)] \cdot (w/p)} \right]^{\frac{1}{1-\alpha}}$$

profit-maximizing # employees: $\ell = [1 + \hat{\tau}(\theta)] n$

$$\ell = \left[\frac{a \alpha f(x)}{w/p} \right]^{\frac{1}{1-\alpha}} \cdot [1 + \hat{\tau}(\theta)]^{1 - \frac{1}{1-\alpha}}$$

$-\alpha / 1-\alpha$

$$\Leftrightarrow \ell = \left[\frac{a \alpha f(x)}{w/p} \right]^{\frac{1}{1-\alpha}} \cdot \left[\frac{1}{1 + \hat{\tau}(\theta)} \right]^{\frac{\alpha}{1-\alpha}}$$